

資料結構

Data Structure

Lab 12

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Lab12-Ex1.

Product Sorting

Code

```
#include<iostream>
#include<fstream>
#include<sstream>
using namespace std;
struct Product{
    string name;
    int day;
    int hot;
};
int main(){
    Product p[100];
    int n;
    ifstream fin("input3.txt");
    fin>>n;
    fin.ignore();
    // 讀取每一個商品
    for(int i=0;i<n;i++){
        string line,word[20];
        int count=0;
        getline(fin,line);
        stringstream ss(line);
        while(ss>>word[count]){
            count++;
        }
        p[i].day=stoi(word[count-2]); // 讀取有效期限
        p[i].hot=stoi(word[count-1]); // 讀取熱銷程度
        p[i].name=word[0];
        for(int j=1;j<count-2;j++){
            p[i].name=p[i].name+" "+word[j];
        }
    }
    // 排順序
    for(int i=0;i<n-1;i++){
        int best=i;
```

```

        for(int j=i+1;j<n;j++){
            if(p[j].day<p[best].day){//比較有效期限
                best=j;
            }else if(p[j].day==p[best].day&& p[j].hot>p[best].hot){//比較熱銷程度
                best=j;
            }
        }
        Product temp=p[i];//排順序
        p[i]=p[best];
        p[best]=temp;
    }
}
//輸出結果
cout<<"商品排序:"<<endl;
for(int i=0;i<n;i++){
    cout<<p[i].name<<endl;
}
return 0;
}

```

Result

The screenshot shows a C++ IDE with the source code for 'Lab12.cpp' on the left and a terminal window on the right. The source code implements a sorting algorithm for a 'Product' struct based on expiration date and sales volume. The terminal window displays the output of the program, showing the sorted list of products: bread, milk, cola, instant noodles, chocolate, and cookie. Below the list, it indicates the process exited after 0.1125 seconds with a return value of 0.

```

C:\Users\User\Desktop\資料結構\Lab12\Lab12.cpp - [Executing] - Dev-C++ 5.11
File Edit Search View Project Execute Tools AStyle Window Help
(globale)
Lab12.cpp
1 #include<iostream>
2 #include<fstream>
3 #include<sstream>
4 using namespace std;
5 struct Product{
6     string name;
7     int day;
8     int hot;
9 };
10 int main(){
11     Product p[100];
12     int n;
13     ifstream fin("input1.txt");
14     fin>>n;
15     fin.ignore();
16     //讀取每一個商品
17     for(int i=0;i<n;i++){
18         string line,word[20];
19         int count=0;
20         getline(fin,line);
21         stringstream ss(line);
22         while(ss>>word[count]){
23             count++;
24         }
25         p[i].day=stoi(word[count-2]);//讀取有效期限
26         p[i].hot=stoi(word[count-1]);//讀取熱銷程度
27         p[i].name=word[0];
28         for(int j=1;j<count-2;j++){
29             p[i].name=p[i].name+" "+word[j];
30         }
31     }
32     //排順序
33     for(int i=0;i<n-1;i++){
34         int best=i;
35         for(int j=i+1;j<n;j++){
36             if(p[j].day<p[best].day){
37                 best=j;
38             }else if(p[j].day==p[best].day&& p[j].hot>p[best].hot){
39                 best=j;
40             }
41         }
42         Product temp=p[i];
43         p[i]=p[best];
44         p[best]=temp;
45     }
46     //輸出結果
47     cout<<"商品排序:"<<endl;
48     for(int i=0;i<n;i++){
49         cout<<p[i].name<<endl;
50     }
51     return 0;
52 }
Line: 25 Col: 40 Sel: 0 Lines: 52 Length: 1245 Insert Done parsing in 0.016 seconds
C:\Users\User\Desktop\資料結構\Lab12\Lab12.exe
商品排序:
bread
milk
cola
instant noodles
chocolate
cookie
Process exited after 0.1125 seconds with return value 0
請按任意鍵繼續 . . .
23°C 多雲時晴 下午 11:22 2026/6/10

```

C:\Users\User\Desktop\資料結構\Lab12\Lab12.exe

商品排序:

bread

milk

cola

instant noodles

chocolate

cookie

Process exited after 0.1125 seconds with return value 0

請按任意鍵繼續 . . .

C:\Users\User\Desktop\資料結構\Lab12\Lab12.cpp - [Executing] - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

IDE-GCC 4.9.2 64-bit Release

Lab12.cpp (globals)

```
1 #include<iostream>
2 #include<fstream>
3 #include<sstream>
4 using namespace std;
5 struct Product{
6     string name;
7     int day;
8     int hot;
9 };
10 int main(){
11     Product p[100];
12     int n;
13     ifstream fin("input2.txt");
14     fin>>n;
15     fin.ignore();
16     //讀取每一個商品
17     for(int i=0;i<n;i++){
18         string line,word[20];
19         int count=0;
20         getline(fin,line);
21         stringstream ss(line);
22         while(ss>>word[count]){
23             count++;
24         }
25         p[i].day=stoi(word[count-2]); //讀取有效期限
26         p[i].hot=stoi(word[count-1]); //讀取熱銷程度
27         p[i].name=word[0];
28         for(int j=1;j<count-2;j++){
29             p[i].name=p[i].name+" "+word[j];
30         }
31     }
32     //排序
33     for(int i=0;i<n-1;i++){
34         int hot=i;
```

C:\Users\User\Desktop\資料結構\Lab12\Lab12.exe

商品排序:

cookies

chips

candy

juice

soda

Process exited after 0.1014 seconds with return value 0

請按任意鍵繼續 . . .

Line: 13 Col: 25 Sel: 0 Lines: 52 Length: 1245 Insert Done parsing in 0.016 seconds

在這裡輸入文字來搜

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C:\Users\User\Desktop\資料結構\Lab12\Lab12.exe

商品排序:

cookies
chips
candy
juice
soda

Process exited after 0.1014 seconds with return value 0

請按任意鍵繼續 . . .

C:\Users\User\Desktop\資料結構\Lab12\Lab12.cpp - [Executing] - Dev-C++ 5.11

File Edit Search View Project Execute Tools AStyle Window Help

TDM-GCC 4.9.2 64-bit Release

(global)

Lab12.cpp

```
1 #include<iostream>
2 #include<fstream>
3 #include<sstream>
4 using namespace std;
5 struct Product{
6     string name;
7     int day;
8     int hot;
9 };
10 int main(){
11     Product p[100];
12     int n;
13     ifstream fin("input3.txt");
14     fin>>n;
15     fin.ignore();
16     //讀取每一個商品
17     for(int i=0;i<n;i++){
18         string line,word[20];
19         int count=0;
20         getline(fin,line);
21         stringstream ss(line);
22         while(ss>>word[count]){
23             count++;
24         }
25         p[i].day=stoi(word[count-2]); //讀取有效期限
26         p[i].hot=stoi(word[count-1]); //讀取熱銷程度
27         p[i].name=word[0];
28         for(int j=1;j<count-2;j++){
29             p[i].name=p[i].name+" "+word[j];
30         }
31     }
32     //排序
33     for(int i=0;i<n-1;i++){
34         int hot=i;
```

C:\Users\User\Desktop\資料結構\Lab12\Lab12.exe

商品排序:

cola
milk
tea
water

Process exited after 0.1075 seconds with return value 0
請按任意鍵繼續 . . .

Line: 13 Col: 25 Sel: 0 Lines: 52 Length: 1245 Insert Done parsing in 0.015 seconds

在這裡輸入文字來搜

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C:\Users\User\Desktop\資料結構\Lab12\Lab12.exe

商品排序:

cola
milk
tea
water

Process exited after 0.1075 seconds with return value 0
請按任意鍵繼續 . . . ■

Discussion

我使用兩層迴圈進行比較，外層迴圈決定要放哪一個位置，內層迴圈找出最適合放在該位置的商品，所以時間複雜度是 $O(n*n)$ 。

我用的演算法是從尚未排序的商品中，找出最應該排在前面的商品，再和目前位置的商品交換，因為我覺得這個做法比較簡單。